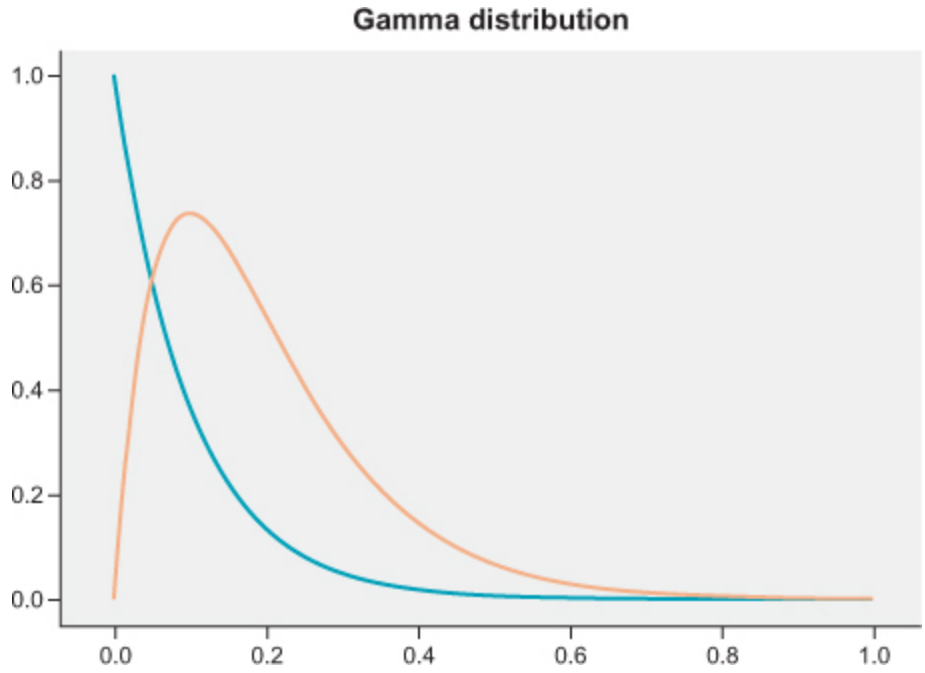
Gamma distribution



Random variables are typically denoted with a capital letter like *X*. In Python, we might set up a random variable using numpy's `random` module:

```
>>> t0 = 18.4
>>> T = lambda: t0 + np.random.randn(1)
>>> T()
array([18.94571853])
>>> T()
array([18.59060686])
```

Here we made *T* an anonymous function that accepts no arguments and just adds a small random number to 18.4 every time it is called. The variance of *T* is 1, which means that most of the values that *T* returns will be within 1 degree of 18.4. If the variance was 10, the spread of likely temperatures would be greater. Generally we start with a prior distribution that has high variance, and as we get more information the variance decreases. However, it is possible for new information to increase the variance of the posterior if the information we get is very unexpected and makes us less certain.

7.3. The Bellman equation